

A study on the impact of pre-processing techniques in Spanish and English text classification over short and large text documents

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Abstract—Nowadays, text mining is a long studied field in science, the vast amount of text resources available has made scientist explore several domains through many different techniques. One of the main processes in text mining are cleaning, reduction and transformation before the application of a classification algorithm. Preprocessing has a large impact in classification algorithms because text is an unstructured form of data with very large number of dimensions and therefore it can be seen as a very sparse matrix. These characteristics that make text so complex are addressed by preprocessing algorithms which extract the main data features. We present a work with a comparison of the performance of the different preprocessing algorithms for a classification problem in two datasets written in Spanish and English.

Index Terms—Text pre-processing, Text classification, Reduction algorithms, Text mining

I. INTRODUCTION

The extraction of meaningful information from texts is a field of science which has gained importance in the last few years, several text tasks can be applied, such as similarity detection, classification, summarization, retrieval, translation, etc. The increase of text resources on social networks, public and private organizations has led to an increase the amount of algorithms and methods to treat and process text data. Text classification is one of the most interesting tasks in text mining, its purpose is to categorize text into a set of different classes and it has wide applications such as email filtering, sentiment analysis, search engines, and digital resource classification such as libraries, news, web pages.

Text is an unstructured form of information with high complexity and a large number of dimensions (words). Working with thousands of dimensions can negatively affect text classification results. Thus, methods which reduce the amount of dimensions can increase the precision and efficiency without losing important features in the data are essential [1]. Such

methods are commonly called text preprocessing and are the main focus of this work.

From the early stages in data mining, data cleaning was part of the mining process. [2] defined the most known data mining methodology (CRISP-DM), in its the "Data Preparation" stage, data cleaning is already taken into account as an important part of the process. A lot of advances since CRISP-DM has came out and subareas in data mining have appeared, thus, specialization in the preprocessing methods of each subarea has born. The text mining field is no exception and due to its complexity, researchers have spent a lot of time working on these. Text preprocessing is one of the most important task given that data in its raw form carries many imperfections, redundancy and inconsistencies[1], in text mining and according to [3], the process to prepare the data it can take up to 80 percent of the total mining process.

There are different preprocessing methods which according to [4] are categorized in three large groups or steps: 1) textual feature extraction based on raw preprocessing of the documents, 2) numerical feature extraction based on term weighting, and 3) dimensionality reduction based on numerical features, also called feature transformation.

The mentioned preprocessing methods treats data differently; 1) raw feature reduction uses morphological aspects in words and simplifies the data by removing not relevant words so called stopwords, processing the roots of words via stemming or converting the words to its most basic forms, and removing unnecessary punctuation and case converting; 2) numerical feature reduction measures the importance of words in the vector space model, one famous technique is document frequency where amount of words are reduced by selecting only the most frequent words considered more important; 3) feature transformation uses transformation methods which measure the words relevance by using different statistic analysis methods, an example is Principal Component Analysis

TABLE I
LITERATURE STUDY

Relevant pre-processing impact literature							
Paper	Raw Feature Extraction	Numerical Feature Extraction	Dimensionality Reduction	Multi Lingual	Short text	Long text	Domain
The impact of preprocessing on text classification [5].	Experimentation	Chi Square		✓	ND	✓	Classification
Evaluating Preprocessing Techniques in Text Categorization [6].	Experimentation	TF/IDF				✓	Classification
Comparing and Combining Dimension Reduction Techniques for Efficient Text Clustering [7].	Fixed	DF - TF/IDF - TfV	LSA - RP - ICA		✓	✓	Clustering
A Comparative Study of Text Preprocessing Techniques for Natural Language Call Routing [4].	Fixed	TF/IDF - TF/GR TF/CW - TF/RF TF/TRR			✓		Classification
This study.	Experimentation	DF - TF/IDF	LSA - PCA KPCA - ISOMAP LLE - NMF	✓	✓	✓	Classification

(PCA) where the words co-variance is measured to select the principal components of a vector.

This work shows several preprocessing methods applied to two datasets and shows the best configuration for every dataset for a classification problem. We use a supervised classification approach and given its popularity and broad application in text mining and its good in text classification [8], [9], we use Support Vector Machine (SVM) classifier and Random Forest (RF) in our classification.

We present an experimental setting where the possible combinations of the type of preprocessing methods were tested, and the main contributions of our work are shown below:

- We compare and show the best possible combinations for two datasets and two languages for the most popular pre-processing methods.
- We compare two classification algorithms and show their difference in the classification results.
- We individually check the results of each type of preprocessing method and also check their behavior in combination with others.
- We analyze the most appropriate number of features and determine the best for each dataset.

This work is organized as follows. Section II, describes the relevant works comparing different preprocessing methods, Section III, shows the different preprocessing methods and methods selected for this work, Section IV presents the performed experiments over different datasets written in Spanish and English belonging to different domains, and finally Section VI presents our conclusions.

II. RELATED WORK

There are several works which study the impact of preprocessing methods over different text mining problems, most of these works focus their efforts on the positive and negative

effects over classification [4], [6], [10], [5] and text clustering problems [7].

Some studies take into account preprocessing methods where raw feature extraction methods such as stop-word removal, stemming, tokenization, and case conversion are applied [5]. Other approaches such as [6], [11], mainly check preprocessing methods based on raw data treatment and feature extraction as TF/IDF but leave feature transformation out of their comparisons. However, we define the preprocessing methods as all the available methods and algorithms applied to the data before a classification or clustering algorithm is applied, these include raw feature extraction, numerical feature extraction and feature transformation or dimensionality reduction, some works which make this differentiation among methods are [7], [4].

Previous studies on the impact of preprocessing methods have approached different preprocessing tasks and algorithms, one relevant work is [4], in which they show the performance of feature selection, in the call routing system domain, and their focus is on supervised and unsupervised weighting methods. This work compares feature selection methods, which take the terms' frequency within documents and a measurement of its importance over the whole dataset. This work takes important weighting methods but it lacks a comparison of different dimensionality reduction approaches. It also fails to tell weather the results would perform differently over different sized documents. In [7] there is a comparison of three numerical feature extraction and three feature transformation algorithms, it compares the performance of Document Frequency, Tf/Idf, Term Frequency Variance as feature selection methods and Latent Semantic Analysis (LSA), Random Projection (RP) and Independent Component Analysis (ICA) as dimensionality reduction methods, on one hand they check each method individually and on the other they combine

some methods to provide insight about a blend performance. However, they leave behind some other important algorithms such as Principal Component Analysis (PCA), Non-Negative Matrix Factorization, and manifold methods such as ISOMAP and Locally Linear Embedding (LLE), also they do - not experiment on the combination with raw preprocessing feature extraction methods into consideration. On the other hand, in [5], they make a comparison of raw feature extraction methods such as tokenization, stop-word removal, case conversion and stemming to later analyze their impact in datasets belonging to two domains written in Turkish and English. However, this work lacks the use of methods such as lemmatization as raw feature extraction method and do not take into account dimensionality reduction.

While it is verified that the classification method greatly influence the resulting accuracy, and [8], [9], [5], [12], [4] say that the best results in text classification of short and long text are given by the method Support Vector Machine (SVM), thus, this work mainly focuses on the impact of text preprocessing methods with the use of SVM as classifier.

The analysis of the relevant work shows important progress done in the field of preprocessing comparison for text classification. However, there is a lack of a complete study showing the best combination of methods which allow to obtain the best possible classification in different datasets written in English and Spanish with different lengths such as short and long texts.

III. PRE-PROCESSING METHODS FOR TEXT CLASSIFICATION

1) *Textual Feature Reduction based on raw preprocessing:* Raw feature reduction methods use the morphology and syntactic aspects into account to perform reductions in the text data. Some of the main methods and those used by this work are described below:

- Stopwords Removal (SP): The SP methods make the assumption that there are some words in a language which do not provide useful information to a text mining process. Words such as articles, conjunctions, prepositions, and determiners are part of these set of words. Therefore, this method removes them from the text.
- Stemming (ST): A stem of a word is its root which caused or originated the word. The stemming method transforms all the words in a text to their stem or root form.
- Lemmatization (LM): A lemma is the canonical form of a word. Thus, the LM method identifies the type of word and transform it to its canonical form. For instance, a verb would be transformed to its infinitive and a noun to its singular form.

2) *Numerical Feature Reduction:* Text mining algorithms take as input a set of numerical features also called vector space model. This can be represented as a numerical matrix where the documents and terms are numerically represented. For this work, we present the following two methods to parse a text document to a numerical vector representation.

- Document Frequency (DF): This is the most basic way to parse a document to the vector space model. This method

counts the frequency of all terms in a document and builds the term-document matrix with all the frequencies.

- Term Frequency / Inverse Document Frequency (TF/IDF): This is one of the most popular methods to reduce a text to its most important features. TF/IDF measures the importance of a term inside a document as well as its importance among the complete set of documents.

3) *Dimensionality Reduction:* There are several studies on algorithms for dimensionality reduction, [13], [14] in their surveys, show some of the most relevant for text mining applications, The following can be named among the most popular: Latent Semantic Analysis (LSA), Principal Component Analysis (PCA), Kernel PCA (KPCA), Nonnegative Matrix Factorization (NMF), Locally Linear Embedding (LLE), among others. Furthermore, [14] concludes that algorithms which analyses on manifolds such as LLE are clearly the most widely used due to their structure adaptation capabilities, also PCA and its variants are popular, and NMF is getting more attraction. Therefore, for this work we check the performance of PCA, KPCA, ISOMAP, NMF, LLE. Although LSA is not mentioned by [14] in its survey, it is widely used and [13] states that it is one of the main dimensionality reduction algorithms for text mining and information retrieval, thus, it is also included in this work. A brief description of the mentioned algorithms is described below:

- Latent Semantic Analysis (LSA): capture the most relevant features in our dataset and by reducing its dimensionality, an example of this can be seen in [15], where Singular Value Decomposition (SVD) is applied to filter out the least relevant features.
- Principal Component Analysis (PCA): PCA analyzes the co-variance of the eigenvectors of the term-document matrix. The algorithms searches for orthogonality components which explains as much variance of the data as possible [14].
- Kernel PCA (KPCA): It is an expansion of PCA which tries to find non-linear relationships. It mainly differs from PCA because it uses a function Φ to transform the input vectors; it can be seen as a dimension expansion as shown in [14].
- ISOMAP: This method aims to find non-linear structures not visible to PCA and MDS. It finds manifolds by "preserving the intrinsic geometry of the data and capturing the manifold distances between all pair of points" [16].
- Non-Negative Matrix Factorization (NMF): This method considers that a drawback of other methods is that they produce matrices with negative values. Thus, this method decompose a dictionary based on the weighted sum of the linear combination given by the topics [14].
- Locally linear embedding (LLE): This methods looks for a manifold close to the data by looking for the k-nearest neighbors of each observation and producing a set of minimized weights for each approximation [14].

IV. EXPERIMENTAL SETTINGS

Text mining approaches usually describe the type of data which works more efficiently, [17], [18] describe topic modeling and classification over short texts and [19], [20] describe text classification and network representation over large corpora or rich text information. There is a clear difference on the classification algorithm behavior for long and short text given the existing number of dimensions of each one, thus, some dimensionality reductions methods that may improve the precision, recall and f-measure results in one category, may affect negatively the other. These experiments present an empirical evaluation of a combination set of the methods shown in section III. For the current work we perform all the different combinations of such preprocessing methods in two different datasets written in English and Spanish.

A. Dataset

For the experimentation two labeled datasets written in Spanish and English were selected. We choose to use as a short text corpus a dataset coming from a micro blogging social network such as Twitter and a long text labeled dataset of syllabi of an Ecuadorian university. It is important to note here that we define as short text as a data corpus whose texts do not have more than 300 words and on the other hand a long text those which have more than 300 words. The datasets are further described below:

- Short Text Dataset: Twitter is a micro blogging social network which has several millions of users. A Twitter policy is to restrict the number of characters in a tweet to 280. Therefore, several millions of users giving opinions over a topic in different languages every day creates an ideal scenario to test the pre-processing methods. We choose automobile transit as domain for our study given that people have diverse opinions about it and offers a chance for a binomial classification through a sentiment analysis. We manually classified 1000 tweets in Spanish and the same amount in English with a positive or negative label.
- Long Text Dataset: As a long text size we use higher education syllabi, in a previous work we classified them in 8 categories described by the UNESCO areas of knowledge. However, this information was available only in Spanish, to overcome this limitation we used a translator service as it is done in [21]. We manually checked a sample of the translated documents and accepted the correctness of the corpus. There are 1442 syllabi in the dataset belonging to all the courses from all the specialties of one Ecuadorian higher education institution.

B. Features Reduction Combinations

In our analysis we use a combination of five different combinations of raw pre-processing methods plus numerical and transformation approaches. We have selected some methods that can be used together without affecting the operation of the others. However, we also include in our experiment setting a case when no raw feature, numerical feature or feature

transformation technique is selected. The following are the mix of raw methods:

- 1) No raw Feature technique (None).
- 2) Stopwords removal (SP).
- 3) Porter Stemmer (ST).
- 4) Lemmatization (LM).
- 5) Stopwords removal + Porter Stemmer (SP+ST).
- 6) Stopwords Removal + Lemmatization (SP+LM).

The following are the Numerical Features methods used in our experiments:

- 1) DF, with this method we keep all the features of the dataset.
- 2) TF/IDF, we selected a threshold of 75% to keep the most relevant features.

The last variable of our study is a feature transformation technique, as it is described in III, there is a large number of these methods. We have selected seven of them discussed in Section III for our study. These methods take as input parameter the number of desire features. Thus, as it is done in [5], we have selected 5 different number of features for each dataset, we used 50, 100, 200, 500, and 750 for both datasets.

As a result of the number of raw (6), numerical (2) and transformation features (7), we obtain 84 possible combinations that are analyzed for both datasets in English and Spanish.

C. Classification Algorithm

We selected two algorithms to perform the classification, Random Forest (RF) and Support Vector Machine (SVM). They are both efficient for text classification and present the best classification results [22], [5]. To measure the efficiency of the algorithms we show the precision, recall and f-measure. We pay more attention to the f-measure as done in [5] given that it represents the combination of precision and recall, thus, it is the most significant measure we have.

V. RESULTS

Section IV stated that there are 84 possible combinations. Therefore, given the large amount of insights that could be reason from the experiments, we first analyze the results from the most general to the most specific.

Figure 1, shows that the f-measure of the classifiers for short texts are better than the those for long texts. This can be explained by the text magnitude, the larger the number of features, the more complex is for the algorithm to correctly classify the different labels. It can also be seen in the figure that SVM performs better when classifying text in both, short and long texts. The medians for short and long text have a better f-measure and it is clear that the 2dn, 3rd and 4th quartiles are higher in SVM than in RF. Thus, we can say that SVM outperforms RF when classifying texts. Therefore, for the next analysis of this study, the results of only SVM algorithm are analyzed.

Figure 2 shows the medians of each pre-processing techniques, we report the median because we checked the normality for every group of data and found out a non-normal

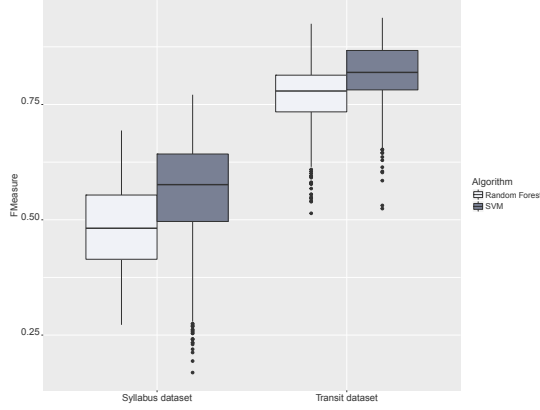


Fig. 1. F-measure results per algorithm

distribution. The type of data (long and short texts) as well as the language (Spanish and English) are split into different analysis, given the purpose of this work.

TABLE II
LEVENE TEST FOR HOMOSCEDASTICITY

Text Type	Type of pre-processing technique		
	Raw	Numerical	Transformation
Syllabus dataset English	0.7921	$3.4E-05$	$1.5E-07$
Syllabus dataset Spanish	0.9077	$1.6E-09$	$1.8E-09$
Transit dataset English	0.1587	$8.9E-14$	$2.2E-15$
Transit dataset Spanish	0.0048	$3.0E-07$	$2.2E-16$

Table II, show the homoscedastic validation assumption. As it can be seen in the table all the values in the raw methods except the transit dataset in English fulfill this assumption. The rest of the values have a heteroscedastic shape, showing that the variance is different. Therefore, for the homoscedastic data we use the Kruskal Wallis's test and for the heteroscedastic ones, we use the Mood's test to check differences in the medians.

TABLE III
PRE-PROCESSING TECHNIQUES SIGNIFICANCE

Text Type	Type of pre-processing technique		
	Raw	Numerical	Transformation
Syllabus dataset English	0.9559	$1.6E-11$	$6.9E-03$
Syllabus dataset Spanish	0.9202	$2.6E-08$	$2.2E-16$
Transit dataset English	0.1307	$1.3E-06$	$2.2E-16$
Transit dataset Spanish	0.4799	$1.9E-08$	$3.1E-15$

Figure 2a shows the raw feature reduction median behavior, A clear trend is shown where the f-measure results of the transit dataset (top part of figure) and the syllabi dataset (bottom part of figure) are clearly different; Short text classification presents a better f-measure than long text. In the top part it can be seen that in every raw reduction method, the f-measure results have are better in Spanish than those in English. Table III shows that for the results of every dataset have no significant differences. There is no statistical difference that

shows that the use of a raw feature reduction method improves the classification.

TABLE IV
BEST RAW FEATURE REDUCTION METHODS BY DATASET

Dataset	Method	Value
Syllabus dataset English	SP + LM	0.83
Syllabus dataset Spanish	SP	0.80
Transit dataset English	No method	0.58
Transit dataset Spanish	LM	0.56

Table IV shows the best medians found in the classification results by raw feature reduction methods. The application of stopwords removal works better for the syllabus dataset; while for the transit dataset there is no a method which is similar to English and Spanish. However, for the Spanish one the application of lemmatization is advised.

Figure 2b shows the medians of numerical reductions, as Figure 2a, it shows the same trend in long and short texts, where short text classification outperforms long text one. However, it shows an interesting trend which make a difference between the syllabus and transit datasets. The application of TF/IDF over a dataset with short texts decreases the classification f-measure while it happens to be the opposite over long texts. The application of TF/IDF in the syllabus dataset increases the overall performance of the classification. A possible explanation for this is that the TF/IDF works better when richer texts are present, this method measures the importance of a term in document better when the text is long. In short texts, this method can fail given that most of the terms would occur just once. Table III shows that the p-values in the numerical features reduction are statistically different.

TABLE V
BEST NUMERICAL FEATURE REDUCTION METHODS BY DATASET

Dataset	Method	Value
Syllabus dataset English	TF/IDF	0.5783
Syllabus dataset Spanish	TF/IDF	0.5819
Transit dataset English	DF	0.8059
Transit dataset Spanish	DF	0.8275

Table V shows the best median values for the numerical feature reduction methods. It can be see in the table that the trend shown in Figure 2b is kept.

Figure 2c shows the feature transformation algorithm medians, as figures 2a and 2b, it shows that these algorithms also have a better f-measure according if the texts are large or short. It is difficult to determine by this graph if there is an improvement with one algorithm or the other. However, it is clear that with the NMF algorithm, the results are poor in comparison with the other algorithms over the transit and syllabus dataset. As it can be seen in table III all the p-values for every dataset for feature transformation are lower than our significance level of 0.05. Thus, the use of a Feature Transformation method does have an effect on the classification results.

Figure 2c shows that the best medians are those of the KPCA, LSA, PCA and None methods. However, Table VI

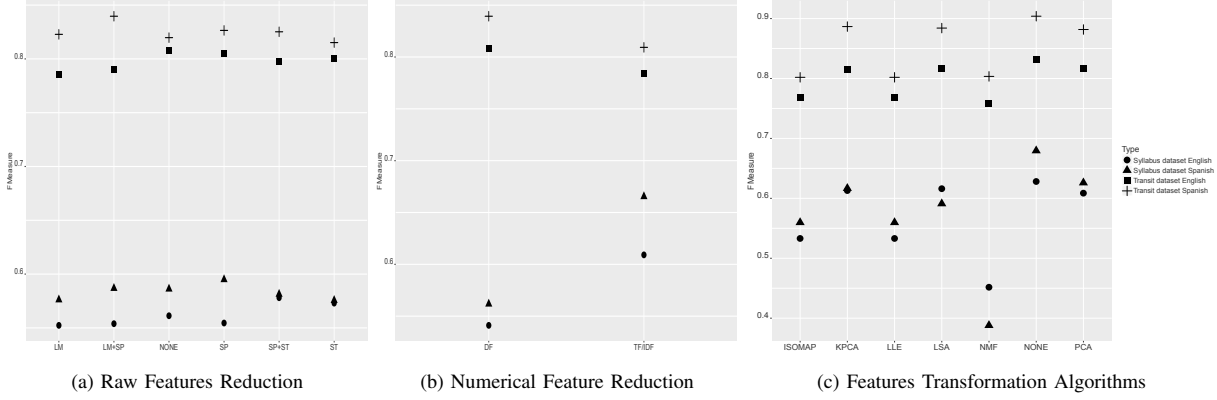


Fig. 2. Classification Results according to the different reduction methods.

TABLE VI
BEST FEATURE TRANSFORMATION METHODS BY DATASET

Dataset	Method	Value
Syllabus dataset English	None	0.6281
Syllabus dataset Spanish	None	0.6795
Transit dataset English	None	0.8319
Transit dataset Spanish	None	0.9039

shows that the best medians for every dataset is when there is no application of a Feature Transformation methods.

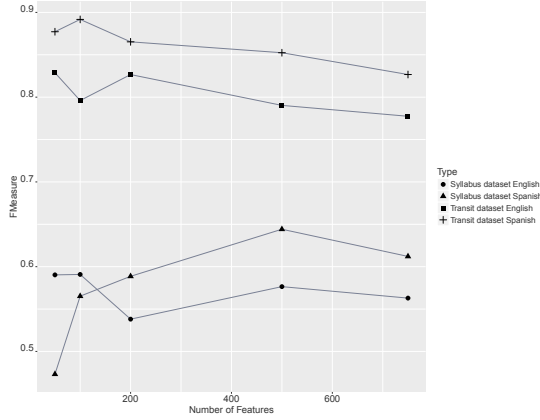


Fig. 3. F-measure per Feature Transformation Dimensions

Figure 3 shows the f-measure medians to illustrate the performance of feature transformation algorithms over a different number of dimensions. As stated in section IV, we selected a number of 50, 100, 200, 500, 750 for our experiments. The behavior for short and long texts present a f-measure increment until a breaking point in which the behavior changes. Transit dataset best performance is given when the number of features is 50 for short texts in English and 100 for short texts in Spanish. In the other hand, for long texts the best performances are given when the number of features is 500. We assume these differences between datasets are given by the number of available terms in each. Thus, the lower the total number of

features in a dataset, the lower dimensions are necessary to achieve the best performances.

TABLE VII
SIGNIFICANCE WITHIN FEATURE TRANSFORMATION METHODS

	Type of Text			
	Long Text 500 features		Short Text 50 features	
	English	Spanish	English	Spanish
Feature Transformation p-value	$6.5E-05$	0.00713	$3.1E-08$	$6.6E-09$

Table VII shows the p-values corresponding to the Mood's test. The values in this tables are lower than our significance value of 0.05. Therefore, the differences among the number of features are significant.

TABLE VIII
BEST POSSIBLE COMBINATION BY DATASET

Dataset	Method			
	Raw	Numerical	Transformation	f-measure
Syllabus dataset English	LM	TF/IDF	KPCA	0.7234
Syllabus dataset Spanish	LM	TF/IDF	None	0.7711
Transit dataset English	SP + ST	DF	None	0.8654
Transit dataset Spanish	SP	DF	None	0.9379

Table VIII shows combinations which got the best results in every dataset from all the 84 combinations described in section IV. It is shown that for the syllabus dataset in English and Spanish the Raw and numerical feature methods are the same, which show that Lemmatization and TF/IDF are the best. However for feature transformation there is not common method. For the transit dataset it is shown that the application of stopwords, TF/IDF and no application of feature transformation method is common for both languages.

VI. CONCLUSIONS

According to our study the following conclusions can be done:

- There is not statistical evidence to show that the application of raw feature reduction methods such as Stopwords,

Lemmatization and Stemming does affect the outcome of SVM in the syllabus and transit datasets. However, it is suggested to apply methods such as stopwords in long datasets and lemmatization in small datasets.

- The application of a numerical feature reduction such as TF/IDF do have an impact in the classification results in long and short text. It positively affects the performance of the classification when the text is long increasing the f-measure. However, when the text is short it is preferable to not use any numerical feature reduction, the use of a count vectorizer shows the best results.
- Feature transformation techniques do have an impact in the classification accuracy. However, it is shown by our results that it is more convenient when using the SVM algorithm to not use any method.
- The selection of the number of features for a transformation method does make a difference in the classification results. For long texts such as the syllabus dataset, the best number of features is located at 500, while in shorter texts such as the transit dataset, it is located between 50 and 100. This tell us that the longer the dataset more features are needed to perform a better classification.
- The results show that the best methods to perform feature transformation are KPCA, LSA, and PCA. However, our results show that the best classification is given when no method to perform feature transformation is applied.
- For long datasets such as the syllabus one, it is advised to use lemmatization and TF/IDF methods in combination with KPCA, LSA, PCA or no method for feature transformation given that they show the best classification results.
- For short datasets such as the transit one, it is advised to use stopwords removal with a count vectorizer and no method for feature transformation. It is shown that the reduction of dimensionality is not advised for short text datasets given that the dataset is already short and that a classification algorithm such as SVM performs better with more dimension.

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